

Introduction

Molecular energy calculations define molecular conformations, stability, and potential energy surfaces. This is important in computational chemistry because many chemical and biological processes depend on small energy differences, such as whether one molecular conformation is more stable than another. Traditional quantum chemistry methods like density functional theory (DFT) provide accurate molecular energies, but the time and cost increase in complexity as the molecular size and dataset size increase. The goal of this project was to reproduce DFT-level molecular energies at much lower computational cost and train a neural network potential following the approach of the ANI-1 paper by Smith, Isayev, and Roitberg [1].

One approach that addresses this limitation is ANI, for ANAKIN-ME, which is a model based on atomic environment vectors (AEVs) as the molecular representation. Instead of feeding the whole molecule into one neural network, ANI builds a local Atomic Environment Vector (AEV) for each atom and feeds it to a neural network specific to that element [2]. The model predicts the atomic energy contributions, which are summed to give an estimate of the total molecular energy. This structure is useful because it keeps the model transferable across molecules with different numbers of atoms while still preserving information about the local chemical environment [2].

In this project, I implemented a smaller ANI-style model using TorchANI and PyTorch. The model was trained on the s01-s04 subset of the ANI/GDB dataset, containing organic molecules with H, C, N, and O atom types. Compared with the full ANI-1 paper, this project used a much smaller dataset, a simpler neural network architecture, and fewer training epochs because of limited computational resources. The main focus was to build the full training pipeline, test different hyperparameters and model architectures, and evaluate whether a simplified version of the ANI approach could still predict molecular energies accurately. Model performance was measured using mean absolute error (MAE) in kcal/mol, and the final results were compared with the original ANI-1 paper to understand how well this smaller implementation reproduced DFT-level energy prediction at lower computational cost.

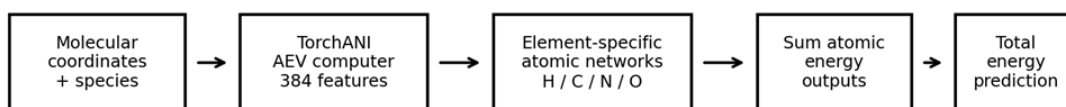


Figure 1. Workflow of checkpoints 1-5; molecular coordinates and species are converted to AEVs, evaluated by element-specific atomic networks, and summed into a total molecular energy prediction

Methods

The dataset used in this project was loaded from the ani_gdb_s01_to_s04.h5 file, which contains organic molecules made up of hydrogen, carbon, nitrogen, and oxygen atoms. The S01-S04 subset contains approximately 81,920 total conformations, with each molecule appearing multiple times across different geometries. The 80/10/10 split yielded approximately 65,536 training conformations, 8,192 validation conformations, and 8,192 test conformations, with each conformation containing up to 14 atoms. Before training, the dataset was preprocessed by subtracting self energies to isolate the interaction energy contribution from each conformation, converting atom symbols to species indices in the order H, C, N and O, and shuffling the molecules. Data for Checkpoints 3 and 4 were split into 80%, 10%, and 10% training, validation, and testing sets. For Checkpoint 5, the best selected model was retrained using 100% of the available s01-s04 data (Figure 1). This means that the Checkpoint 5 value is a training MAE, not a held-out test MAE.

The model used atomic environment vectors, or AEVs, to describe the local chemical environment of each atom. These vectors were generated using radial and angular symmetry functions with cutoff parameters of $R_{cr} = 5.2 \text{ \AA}$ for the radial term and $R_{ca} = 3.5 \text{ \AA}$ for the angular term, using 16 radial shifts and $8 \text{ angular} \times 4 \text{ radial}$ shifts in the angular term. The AEV dimension used was 384. This representation served the same general purpose as the AEVs in the original ANI-1 paper, which was to convert each atom's surrounding chemical environment into a numerical input that could be passed into an element-specific neural network [2].

The architecture used a separate neural network for each atom type of H, C, N, and O similar to the ANI model setup. Each atomic network took an AEV as input and predicted one atomic energy contribution, and the atomic energy contributions were then summed by torchani.ANIModel to predict the total molecular energy. The torchani.ANIModel was helpful since it applied the correct atomic network to each atom based on its element type and then combined the atomic outputs into one molecular energy prediction. In the final implementation, the AEV computer and ANI model were combined in an nn.Sequential model. This made the workflow cleaner because the model could take species and coordinates as input, generate AEVs, and then pass those AEVs into the ANI model for energy prediction. The model that had the best performance used one hidden layer with 128 units, ReLU activation, and dropout of 0.1, giving a total of 197,636 trainable parameters.

Training was performed using the Adam optimizer and mean squared error loss in Hartree units. Training was conducted in batches of 128 conformations, and each configuration in the hyperparameter search was trained for 10 epochs. Model performance was assessed using mean absolute error in kcal/mol, using the conversion that 1 Hartree is equal to 627.509 kcal/mol. During hyperparameter tuning, a new model was initialized for each trial so that each configuration could be compared without continuing training from a previous trial. This was important because reusing a previously trained model would make the hyperparameter comparisons biased.

Results

The best overall performance came from the simplest architecture tested in the hyperparameter search. Figure 2 summarizes all configurations evaluated during the hyperparameter search. The best-performing model used one hidden layer with 128 units, a learning rate of $1e-4$, L2 regularization of $1e-5$, and dropout of 0.1. This model achieved a held-out test MAE of 1.21 kcal/mol, which was below the target of 2 kcal/mol. Overall, the models trained with a lower learning rate performed better than those trained with the $1e-3$ learning rate, but all tested models were below the 2 kcal/mol threshold.

Hidden layers	LR	Epochs	L2	Dropout	Train MAE	Val MAE	Test MAE
[128]	$1e-4$	10	$1e-5$	0.1	1.22	1.23	1.21
[128]	$1e-4$	10	$1e-5$	0.0	1.24	1.24	1.23
[256, 128]	$1e-4$	10	$1e-5$	0.0	1.33	1.34	1.32
[256, 128]	$1e-4$	10	$1e-5$	0.1	1.36	1.37	1.35
[256, 128, 64]	$1e-4$	10	$1e-5$	0.0	1.49	1.50	1.48
[256, 128, 64]	$1e-4$	10	$1e-5$	0.1	1.52	1.53	1.52
[128]	$1e-3$	10	$1e-5$	0.1	1.56	1.57	1.56
[256, 128]	$1e-3$	10	$1e-5$	0.0	1.57	1.59	1.57
[256, 128]	$1e-3$	10	$1e-5$	0.1	1.63	1.65	1.63
[128]	$1e-3$	10	$1e-5$	0.0	1.68	1.69	1.67
[256, 128, 64]	$1e-3$	10	$1e-5$	0.1	1.78	1.80	1.78
[256, 128, 64]	$1e-3$	10	$1e-5$	0.0	1.83	1.84	1.83

Figure 2. Hyperparameter tuning results for the AEV-based energy prediction model. The best-performing model used one hidden layer with 128 units, a learning rate of $1e-4$, L2 regularization of $1e-5$, and dropout of 0.1, reaching a held-out test MAE of 1.21 kcal/mol.

The hyperparameter results clearly showed that the model performance does not improve with increased depth for this dataset and training setup. The models with two hidden layers or three hidden layers had higher errors than the models with one hidden layer. The lower learning rate of 1×10^{-4} likely allowed the optimizer to navigate the loss landscape more carefully within the 10-epoch budget, avoiding the overshooting that appears to affect the 1×10^{-3} configurations. With the limited training time, this indicates that adding more layers did not improve generalization for the smaller s01–s04 dataset. This suggests that the deeper models may have needed more training time, more tuning, or a different regularization setup to outperform the smaller network. Since GPU time was limited, the smaller one hidden layer model was the most practical and most accurate option.

After identifying the best hyperparameters, I repeated training to evaluate the stability of the result across multiple training runs and across five runs, the model reached a mean test MAE of 1.245 ± 0.082 kcal/mol. This was close to the best single run result, which suggests that the selected architecture was reasonably stable across random initialization and data shuffling.

Overall, the multiple runs with the optimal hyperparameters show consistency in mean test MAE, which means that the best result was not due to a favorable training run. After retraining and evaluating the best model, the learning curve showed that the training loss decreased sharply during the first few epochs, then continued to decrease more gradually (Figure 3). This suggests that the model learned most of the structure-energy relationship early in training. The validation MAE stayed relatively stable across epochs, with only a slight downward trend, indicating that the model was not strongly overfitting during this run.

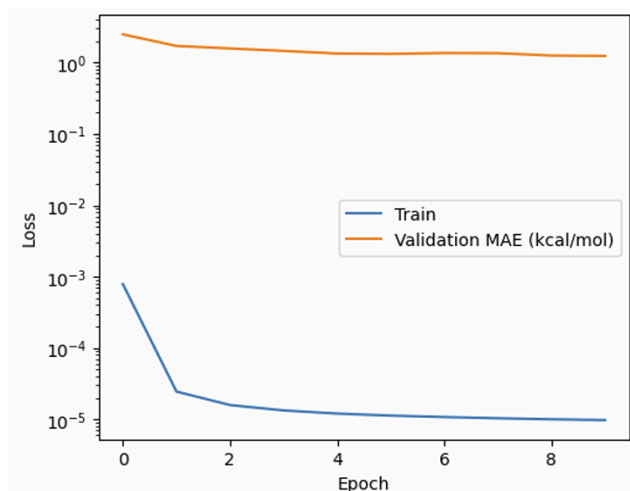


Figure 3. Learning curve for the selected AEV-based energy prediction model - The training loss decreased rapidly in the first epochs and then began to level off, while the validation MAE stayed relatively stable throughout training.

I also evaluated the selected model using 5-fold cross-validation. The mean test MAE from cross-validation was 1.455 ± 0.209 kcal/mol, which was higher than the repeated-run average, indicating that model performance depended on the train/test split. However, the average cross-validation performance still remained below the 2 kcal/mol target, supporting that the simplified ANI model was able to learn useful structure-energy relationships from the dataset.

For the final checkpoint, the selected model was retrained on 100% of the available s01–s04 data for 30 epochs. This produced a training MAE of 1.239 kcal/mol. Because the final model was trained on all available data, this value should be interpreted as a training result rather than a held-out test result. The parity plot showed strong agreement between the predicted energies and the reference DFT energies. Most points fell close to the $y = x$ line, indicating that the model predictions closely followed the reference labels. The remaining spread around the line represents prediction error, with somewhat larger deviations at the higher-energy end of the plot. This pattern is reasonable because higher-energy conformations are often less common and structurally more complex, making them harder for the model to predict accurately [3]. Overall, the model learned a strong relationship between AEV-based molecular structure and total energy.

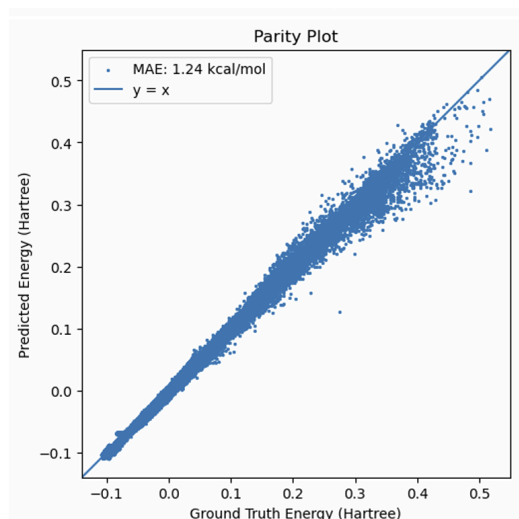


Figure 4. Predicted versus reference DFT energies for the final model. The predicted molecular energies closely followed the reference DFT energies, with most points clustered near the $y = x$ line.

Compared with the original ANI-1 paper, this project achieved a higher error, which was expected because this implementation used a smaller dataset, a simpler network, and more limited computational resources. The original ANI-1 model was trained on around 17 million conformations with a larger symmetry function set that produced a more diverse set of AEVs and was trained for significantly more epochs, which contributed to a lower generalization error. The scale difference is also reflected in model complexity, as the original ANI-1 network used approximately 2.5 million trainable parameters per element network, compared to the 197,636 total trainable parameters used in all element networks. The ANI-1 paper reported a test MAE of approximately 0.57 kcal/mol [2], while the best held-out test MAE from this project was 1.21 kcal/mol. Although this model did not match the full ANI-1 result, it still demonstrated the central idea of the ANI method that atomic environment vectors combined with element-specific neural networks can predict molecular energies with meaningful accuracy.

	Architecture	Data Used	Epochs	Reported Metric
Model / Run				
ANI-1 Paper	deeper ANI model	full ANI-1 dataset	many more than this project	0.57 test MAE
Checkpoint 3/4 best	1 hidden layer [128], ReLU	s01-s04, 80% train / 10% val / 10% test	10	1.21 test MAE
Checkpoint 4 multi-run	1 hidden layer [128], ReLU	s01-s04, 80% train / 10% val / 10% test	20	1.245 $\pm$ 0.082 test MAE
Checkpoint 4 k-fold	1 hidden layer [128], ReLU	s01-s04, different folds	10	1.455 $\pm$ 0.209 test MAE
Checkpoint 5 all-data final	1 hidden layer [128], ReLU	s01-s04, 100% train	30	1.239 training MAE

Figure 5. Comparison of project results with the ANI-1 paper: This project's best held-out test MAE was 1.21 kcal/mol. Although higher than ANI-1, this result shows that a simpler model trained on the smaller s01–s04 dataset can still achieve useful molecular energy predictions.

Discussion

This project successfully implemented a simplified ANI-style neural network potential for molecular energy prediction using the s01–s04 subset of the ANI/GDB dataset. By combining atomic environment vectors with separate element-specific neural networks, the model learned the relationship between local atomic structure and molecular energy while preserving the main idea behind the original ANI approach. The best held-out model achieved a test MAE of 1.21 kcal/mol, meeting the project target of below 2 kcal/mol and demonstrating that the approach can still produce accurate molecular energy predictions at a smaller scale. Five-fold cross-validation confirmed the robustness of this result, yielding a mean test MAE of 1.455 ± 0.209 kcal/mol.

The hyperparameter search showed that a simpler one-hidden-layer model performed better than deeper alternatives, indicating that learning rate and regularization choices had a larger impact on performance than adding more layers. Although the model did not match ANI-1's accuracy, that difference is expected because ANI-1 used a much larger dataset and more computational resources, such as training time, number of trainable parameters, and more conformations. Overall, the results show that the ANI framework can be adapted successfully for molecular energy prediction with a smaller subset of data. Performance can be improved by training on a larger portion of the ANI dataset, using longer training runs with a learning-rate schedule, testing larger element-specific networks, and analyzing errors by molecule size, atom composition, or energy range. It would also be useful to evaluate the model on molecules outside the s01–s04 subset to better test transferability.

References

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